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Student	Brooke Struyk
Project	KidsOR + Public Intervention Anesthetic Waste Gas
Role(s)	Team Lead and Design Engineer
Mentor(s)	Dr. Robert Read, Dave Tipping, Courtney Luddick, Dr. Robert Stone
Team	Brooke Struyk Kaayla Osborne Rachel Mensah Sarah Krishnamoorthy
Instructor(s)	Dr. Christopher Arena, Dr. Robert Stone, Dr. Ashley Taylor
TA(s)	Evan Diacoumis, Leah Van Gelder, Ali Stibler

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Timesheet

Table 001-1 Timesheet

Name	Week	Hours Worked Including Class	Responsibilities	Student Signature	Instructor/ TA Initials
Brooke Struyk	1	4.5 hours total (2.5 hours in-class; 2 hours outside of class)	• Attended lectures on 8/25 and 8/27 (2.5 hours) and was assigned a team and project. • Completed project selection survey (2 hours).	BHS	
	2	6.5 hours total (2.5 hours in-class; 4 hours outside of class)	• Attended lectures on 9/1 and 9/3 and found a time to meet stakeholders and weekly group meetings (2.5 hours). • Completed reading reflection map assignment (1.5 hours). • Completed theoretical or numerical analysis assignment (2 hours). • Did initial research on the project by going through the project's linked GitHub (0.5 hour).	BHS	
	3	10 hours total (2.5 hours in-class; 7.5 outside of class)	• Attended lectures on 9/8 and 9/10 (2.5 hours). • Had an initial team meeting on 9/9 to prepare for stakeholder meeting, create roles, 60 second project pitch, find interviewees, etc. (1.5 hours). • Met with stakeholder on 9/10 and asked questions about project (1 hour).	BHS	

		- Completed HOQ assignment (1 hour). - Completed engineering notebook #1 (4 hours).		
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“As a Hokie, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do.”

Summary of Individual Effort

Accomplishments

- What did you get done over the past 2-3 weeks? Progress on action items?
- A good target is 3 meaningful accomplishments per week.
- What helped you move forward?

In the past 2-3 weeks I have done some initial research on how waste-gas machines work, and I was able to help the team come up with questions for when we meet KidsOR. Our group also decided on a day and time for our weekly team meeting. We were able to find common ground, decide roles, and start to get excited about the project. We were able to split up future work for DH01, and we started to create a timeline and plan for interviews and prototype building. During our conversation about what we should ask our stakeholder, we were able to pick each other's brains about initial ideas and potential solutions. This helped us to start to brainstorm ideas and potential future directions. In week 3, we had our first stakeholder meeting, and we were able to learn a lot more about the project. We now have a better understanding of our stakeholders' expectations and our list of questions for KidsOR has increased.

Action Items

- What are you personally responsible for over the next 2-3 weeks and what are the due dates?
- A good target is 3 meaningful action items per week.
- Use terminology that is Specific, Measurable, Achievable, Relevant, Time-bound (SMART)

In the next two weeks, I am responsible for the following tasks:

- I plan to have my interview this upcoming week, so we can understand the general workflow of a pediatric anesthesiologist. I want to have this meeting with this stakeholder after our KidsOR meeting, so we can use it as a comparison of workflow (like the job map assignment). I think it could be interesting to see the differences and similarities in a hospital setting versus the non-profit setting specifically for the anesthesiologist.
- I plan to help hold a meeting with KidsOR and hopefully I can attend, so that I can use that information to ask my own questions and come up with new ideas on our potential solution.
- I plan to complete my sections of DH01 over the weekend, so during our meeting on Wednesday (9/16), we are able to fully review each other's sections.

Personal Challenges and Professional Growth (self-awareness)

- What have you found to be technically challenging? What risks are ahead?
- Explain how you are trying to grow as a professional and engineer in the project.

The beginning of the project was a bit rocky. I honestly felt that every time I came to class, we were already a week behind not doing enough. This fear came from having first stakeholder meeting on week 3 on a Thursday. I am lucky that my interviewee is easily accessible because he is my parent; however, this means we would have a week and a half to get 3 more interviews for our assignment due 9/22. After asking other groups I realized we were all in a similar situation and my stress was not helpful for the group. Our first team meeting was also helpful because we were able to gain clarity on deadlines, make interview questions, and help each other find interviewees. As of right now I feel we are in a good spot, and I am excited to get more information about the project and how to help.

Team Dynamics

Try each of these retrospective tools to learn what works well for your team. Bring your individual responses to the team meetings and make time to discuss and document them as a group.

- **Submission 1 - SWOT**
 - Strengths of team? E.g., leadership, culture, collaboration, inclusion, planning, habits...
 - After our first meeting I would say our biggest strength is collaboration. When we were creating the 60 second project pitch, we were able to bounce ideas from one another, add onto each other's thoughts, and be honest about getting rid of certain lines or not agreeing with certain ideas. In the end our pitch was great,

and we were the first group to raise our hand in class and share. We are also all very qualified technically. We all have different strengths in software, design, electronics, etc., so as of right now it doesn't look like there is any big gap in our project's skills.

- Weaknesses of team?
 - Our biggest weakness is time. When filling out a When2Meet, it became clear that everyone in our group is extremely busy when it comes to classes, clubs, research, and work. Next semester this might be our biggest challenge when we have to collaborate and build out this project. Only two of the group members are able to meet with our main advisor as well. Even with the time constraints, we have a weekly meeting that is a set time when everyone is available, a text group chat, and we are all very good at communication. These weekly meetings will be great for catching up with the group and making sure everyone is aligned.
- Opportunities for the team?
 - One interesting opportunity that was brought up to us is testing our sensors using anesthetic gas. When I first read this project, I was also concerned about the testing phase, but our advisor informed us that he would be able to help us find a place to test (VT Vet Med, Carilion, etc.). This is exciting news, and it will be super interesting to see how our sensors we create compare in a hospital versus veterinary clinic.
- Threats to the team?
 - As of right now our biggest threat is time. There are a lot of assignments due, and everyone is extremely busy. We all have great communication skills, but it almost feels like there needs to be more hours in a day to get everything we need to get done in time. Once we complete the first document for DH01, we will all have a better understanding of expectations for our project, and the other assignments will become easier.
- **Submission 2 - 4 L's**
 - What did you Like?
 - What did you Learn?
 - What did you Lack?
 - What did you Long-for?
- **Submission 3 – KALM**
 - Keep: Something the team is doing well and should continue doing?
 - Add: Something new that the team should incorporate into the next 2-3 weeks?
 - More: Something that brings value to the team, and you should do more of?
 - Less: Something that isn't going as well, and you should do less of?
- **Submission 4 – You pick!**

Additional notes for instructor/TA

- Any general comments or questions?
- Would you like to request a meeting? Please email Dr. Arena (carena@vt.edu).

No comments/questions for this submission.

Individual Meeting Minutes

<Document the team's discussion during formal meetings with advisors, sponsors, subject matter experts (SMEs), or each other, etc. as it relates to your **personal perspectives**. Be reflective and document your own responsibilities, insights, rationales, methods, and analyses. Pay particular attention to laboratory safety (e.g., lab safety practices, product/service safety considerations, and safety regulations related to design).>

<Only the **team leader** should be responsible for providing a summary of what the other **group** members are working on. We DON'T want to see repeated information here across team members.>

Initial Team Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 09/09/2026 6:30pm-8:00pm
- Format: In-person
- Individual notes:
 - Complete DH01 sections by 9/16 for review.
 - Prepare for interview with stakeholders on 9/10.
 - Start to create questions for interview with pediatric anesthesiologist.
- Team Discussion (**team leader** only):
 - We established roles on the team.
 - We were able to come up with over 10 questions for our stakeholder meeting on 9/10.
 - We assigned responsible and supporting roles for DH01
 - We assigned a designated area for where our files will be found (Teams).
 - We worked on finding interviewees for each member.
 - We decided on day and time for weekly team meetings for the entire semester.
 - We created the 60second project pitch for 9/10 class.

Meet the Stakeholders

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy, Dr. Robert Read, Dr. Ashley Taylor, Dr. Robert Stone
- Date and Time: 9/10/2026 3:30pm-4:30pm
- Format: Zoom
- Individual notes:
 - We need to contact the Scotland team and them a lot of questions about the machine and what they are doing to get rid of anesthetic waste-gas.
- Team Discussion (**team leader** only):
 - We reassigned an area for where all our files will be found after talking to our stakeholder (GitHub)
 - We were able to get clear expectations for the project and further understand what needs to be completed.
 - We decided on biweekly meetings with our stakeholder and to increase this to weekly in the spring semester.
 - We were able to discuss initial ideas for the project with our stakeholder.
 - We established our future meeting formats, and our stakeholder is very easy to reach by email.

Literature Review

<It is expected that each team member will read and summarize **three new pieces of literature with each new submission!** Keep track of the total number below by incrementing the header. You can choose from peer-reviewed, scientific articles (<https://scholar.google.com/>), patents (<https://patents.google.com/>), engineering standards (request from Dr. Arena ahead of time if not accessible through VT or ANSI), or FDA guidance documents (<https://www.fda.gov/regulatory-information/search-fda-guidance-documents#guidancesearch>). **Do NOT use literature that was separately assigned for homework.**

Note: AI should serve as a collaborator, not a replacement, for your own critical thought and skills. The intellectual contribution and design decisions must remain your own. AI tools may lack direct access to Google Scholar/Patents and the most recent literature. Additionally, AI-generated content can be inaccurate, biased, or incomplete. Therefore, it is up to you to identify the key pieces of literature for your project. AI can be used to support the Summary of Key Findings section, but you should verify the accuracy of the content and that nothing from a piece of literature was overlooked. Please cite the AI tool used, the prompt that you provided, and your analysis of, or updates to, the content. The other sections should be completed without the use of AI.>

1. Anesthesia and its environmental impact: approaches to minimize exposure to anesthetic gases and reduce waste [1]

Source Authors, Journal, Date, Affiliation

- [1] K. Samad, Muhammad Saad Yousuf, H. Ullah, Syed Shabbir Ahmed, Khalid Maudood Siddiqui, and A. Latif, "Anesthesia and Its Environmental Impact: Approaches to Minimize Exposure to Anesthetic Gases and Reduce Waste," *Medical Gas Research*, vol. 15, no. 1, Jun. 2024, doi: 10.4103/mgr.medgasres-d-23-00059.

Summary of Key Findings

Anesthesia is a mix of nitrous oxides and halogenated ethers that are strong greenhouse gases and contribute to global warming. Operating rooms account for 20-30% of waste leaving the hospital. By improving anesthetic waste gas sensors and collecting systems or changing the flow rate of anesthetic gas, there would be less output. It is important to consider the environmental effect of anesthesia, its contribution to greenhouse gas emissions and global warming.

Application to Design Project

In our initial stakeholder meeting we talked about creating a system that would flow anesthetic gas out of a window. Our stakeholder was unsure what this would mean for the environment if this ended up as a solution, so I decided to use one of my articles to read more about it. The article talks about low-flow anesthesia, so it would help reduce the number of halogenated ethers entering the environment. It would be a good question to ask KidsOR if they are using low-flow anesthesia methods. As of right now the team also uses a chemical bed that collects this fluorine that's bad for the air and saturates it by forming mineral salts, which might be better for the environment already. This is something we can do further research on.

Specific Information for the Design project

- Acute care settings account for 8% of greenhouse gas emissions
- United States healthcare sector contributes to nearly 10% of the total GHG emissions with hospitals being the primary source
- Operating rooms generate 20-30% of the entire waste output of a hospital
- Recommended ways to minimize anesthetic gas waste exposure: LFA, advancing scavenging system, vaporizer, closed-loop delivery system, routine maintenance and calibration, and waste gas monitoring

2. An Initial Evaluation of a Novel Anesthetic Scavenging Interface [2]

Source Authors, Journal, Date, Affiliation

[2] J. A. Barwise, L. J. Lancaster, D. Michaels, J. E. Pope, and J. M. Berry, "An Initial Evaluation of a Novel Anesthetic Scavenging Interface," *Anesthesia & Analgesia*, vol. 113, no. 5, pp. 1064–1067, Nov. 2011, doi: 10.1213/ane.0b013e31822c9a2c.

Summary of Key Findings

The primary method used in the United States and Europe is an active gas scavenging system (AGSS). The machine requires a 24-hour per day period which leads to higher energy costs. By using a DGSS, you do not need a pump as big of a pump, energy costs are lower, and the device can be turned on and off. The article does note that there is no sensor display or failure modes that have been identified.

Application to Design Project

It was brought up during the stakeholder meeting that we should investigate anesthetic scavenging devices as a potential solution, so I decided to read more about it. The article depicts a solution from Vanderbilt University in 2011. If we were to use/ create our own scavenging system this design approach would be a good place to start. The researchers were able to save \$28,000 after a 6-month testing period. Though it would also be \$2738 per year on electricity. We need to figure out what KidsOR is using and how much it is costing them per year, so we could potentially use a similar solution and optimize it. The device built does not have any display or failure mode, so it's not necessarily what we are doing unless we wanted to start from scratch. This idea might also end up costing KidsOR more money than building onto their existing solution of the bed chamber.

Specific Information for the Design project

- Waste anesthetic gas scavenging technology has not changed much in the past 30 years
- Process requires a large vacuum pump that is expensive and operates continuously
- Standards for waste anesthetic gas were recommended but never implemented
- AGSS works by: "Exhausted gas from the anesthesia machine is removed by an open manifold connected to a vacuum source. The system entrains and evacuates waste gas with relief from positive and negative pressure in the breathing circuit. The balance is achieved either with relief valves that open to atmosphere with a slight pressure gradient or valveless 'tube-within-a-tube' system"
- Many hospitals and ambulatory centers used a dedicated scavenging system that has a separate pump and piping.

3. Waste anesthetic gas exposure and strategies for solution [3]

Source Authors, Journal, Date, Affiliation

[3] H.-B. Deng, F.-X. Li, Y.-H. Cai, and S.-Y. Xu, "Waste Anesthetic Gas Exposure and Strategies for Solution," *Journal of Anesthesia*, vol. 32, no. 2, pp. 269–282, Feb. 2018, doi: 10.1007/s00540-018-2448-1.

Summary of Key Findings

The article brought up that different countries have different standards for how to treat anesthetic waste gas. It also brought up that long term exposure can lead to chronic diseases that based on their research harm the anesthesiologist the most later on. This is especially concerning because there is no standard biomarker test for elongated waste gas exposure. The article recommends using a scavenging system, air-conditioning system, Iso-Gard Mask, or passive scavenging system with activated charcoal canister to effectively get rid of gas waste.

Application to Design Project

This article brought up a few interesting points that there is no set limit for the amount of sevoflurane given to a patient, and there is also no universal standard for waste anesthetic gas control. KidsOR travels to different countries and had to abide by different standards, so I think this would be interesting to ask if there is some initial research they have to do before going to a new country and administering anesthesia. The article also brought up concerning biomarkers for WAG exposure after a long period of time. I think it would be interesting to ask the team if they recognize they do any yearly testing for these biomarkers. It would also be interesting to ask my stakeholder, who is a pediatric anesthesiologist, if the hospital he works at has yearly tests.

Specific Information for the Design project

- It's impossible to fully eliminate waste anesthetic gas (WAG) from the air
- Long term exposure to WAG can lead to increased oxidative stress DNA damage, genotoxicity, and carcinogenesis
 - Some have found an increase in chronic diseases, spontaneous abortion, congenital malformations, low birth rates, and higher stillbirth in offspring
- Check out the National Institute of Occupational Health and Safety and American Conference of Governmental Industrial Hygienists for future literature reviews on WAG exposure
- 28% of OR's complied with predetermined standards about WAG
- Different countries have their own limit standards (ADD ABOVE)
 - There is a table on page 3 that would be useful to reference exposure limits in different countries
- There is no set limit for the amount of sevoflurane used
- WAG is higher when patient is in a sitting position compared to standing
- Anesthesiologist is the one getting the most amount of WAG exposure compared to the rest of the OR
- There is no "good" biomarker testing WAG exposure levels

Individual Design Updates

<This section is your electronic design notebook. Include sketches, photos, engineering drawings, equations, calculations, tables, graphs, observations, etc. **Focus on your narration of the design process. For example, your discussion of trainings that you complete and the outputs and/or challenges of that training process (e.g., COMSOL tutorial results, CAD training results, etc.).** Be sure to describe each item in detail with explanatory text so that it is easy to understand when reviewing. It is up to you on how to effectively organize the section headers. Include any notes about updates to User Needs (UN), Design Inputs (DI), Design Outputs (DO), Design Verification (VER), and Design Validation (VAL). **There should be at least 3 new pages with each new submission.** In select cases, you may paste and circle **individual items** from your Mural or Miro team boards and describe your contribution in the text below. However, most of the content in this section should be your **individual work**. Make sure that the font size is large enough to be legible, including within figures. However, **a majority of the 3 pages (≥ 1.5 pages) should be your own descriptive text or narration of the design process. Additionally, it is expected that a majority of the content ($\geq 1/2$) is new or expands beyond what was completed as part of other assignments, such as reading reflections and DHF's. If you include items from a prior individual assignment, please note that in bold/red and expand your discussion.**>

Defining User Needs and Purpose

Questions for 09/10 meeting: - These questions will have to be asked to Scotland team...

1. Talk us through the surgery process.
2. Tell us about the waste management bed and how you interact with it.
3. What specific issues are you facing with the current solution?
4. What types of surgeries typically experience the drowsiness?
5. What type of anesthetic gas is being used?
6. Are there any other chemicals we should consider?
7. At what point in the surgical process do you realize you are drowsy?
8. What are the parameters that control the flowrate of gas?
9. Send pics?
10. What do the rooms for kids OR look like?
11. What models of anesthetic machines do you use?
12. Is there a preventative maintenance procedure for this?
13. Financing about the device? - Ask about a realistic investment in this device

Project pitch:

Imagine you're in surgery and the surgeon falls asleep in the operating room with you. When you think about nurses, doctors, and care technicians you think of you people right there and concentrate the oxygen on top of already exhausted medical personnel, anesthetic gas may escape and cause drowsiness for everyone in the OR. Anesthetic gas is administered to the patient and waste is collected by a tank, known as a chemical absorber bed. The fluorine and chlorine in the gases binds to sodium and chloride in the absorber bed to create a salt. As more gas flows into the bed, saturation increases. There is currently no indicator of when the tank is full and the gas has reached unsafe levels. KidsOR and PublicInvention are global organizations combating care inequities and design gaps in low-resource settings. Together, our project seeks to create a solution for the medical personnel who face this daunting challenge daily.

Meeting Notes:

- 10-20 surgeries a day
- Select days for each surgery
- OR is in literal shipping container. In typical ORs the room in itself has a waste gas filtering system
- 1 in 10 have gas scavenging systems
- Room is the size of a 18 wheeler truck
- In most systems there is exhaust in floor or ceiling
- Open circuit system
 - close to the exhaust is a suction fan, avoiding negative pressure on the lung when connected to the child
- Scrubbing the gases is not very important
- Take the gases out without making a tripping hazard and get it out of the OR
- If it going outside someone outside could potentially get sick (unimportant detail)
- Anesthetic gas is not dangerous for the environment

Dr. Read solution: put a fan next to exhaust, connected to a tube, run the tube on the floor tape it down and it goes out the window, there is a switch

- Should it be the anesthetologist who turns this switch?
- Modularity is important
 - Sensor
 - Source of suction (he wanted a fan next to operating room could be anywhere)
 - The tube is important
- Scavenging system is where we should start
- US\$1c charity -> PublicInvention

-Share design history files with Dr. Read

-Decided on meeting structure

- Bi-weekly meetings
 - Email during weeks (3 by 3 notes - strengths, weaknesses)
 - Loves emails
 - Ask questions
- Add files to GitHub preferred
- Wants everything

Parameters to consider:

- What should a typical air flow be?
- When using a fan CPU ventilation could be enough?
- What typical outflow is?
- Consider the air intake for a child.
- How far would the window or the closest to an exit be?
- Request pictures of the room?
- Sivoflourine used typically for anesthetic
 - What level would it affect a person if breathing for 8 hours?
 - Is this detectable with a inexpensive sensor?

Preventative procedure:

- Ask KidsOR
- Typically install a solar panel system
- Operating independent of the local area
 - Don't have enough power for HVAC unit
- If solution is a fan out it will also be taking out AC from the room and would cost more money objectively
- Turn on automatically in the presence of gases
 - Have a test/maintenance mode
- Can open window/turn up fans
- Ask about ventilation system

-Dr. Stone can provide anesthetic gas for testing

17%

We created a project pitch for class that is on our Miro board. This will be helpful later in the DH01 document describing the purpose of the project and creating later project pitches for presentations. I also added a couple questions to ask the KidsOR team when we meet them so we can understand their current solution and how our eventual solution can fit into what they are doing. There is also a list of the meeting notes from the initial team meeting. These notes were transferred onto a word document and combined with the rest of the group, so we are all clear on expectations and what our stakeholder initially told us. All these additions cover the initial meet the stakeholder portion of the design process. Without keeping track of notes and comments from our stakeholder, we cannot start to create user needs.

Theoretical or Numerical Analysis Homework

4. Explain the range of quantitative input values that you would use and the range of values you would expect for the output. Describe how you determined these estimates.

A potential anesthetic gas sensor that could be used for this project is the MPS4SENS Anesthesia Sensor made by Micro-Hybrid [1]. I found this by looking for anesthetic gas sensor datasheets, so I would be able to find an example voltage input value. This device specifically measures CO₂, N₂O, desflurane, and isoflurane. In the specifications, the listed DC output voltage is around 9.35mV. This value can be used as an example quantitative input value for voltage. Since this is a small voltage, a non-inverted amplifier could be used for the Arduino to read the signals. Assuming the Arduino input range is between 0V and 5V, I'm using 2.5V as a voltage output target. In this case, you would calculate $A = V_{out} / V_{in} = 2.5 / 0.00935 = 267$. To calculate the resistor values for a non-inverting amplifier, $A = 1 + R_1/R_2$. The gain needs to be $R_1/R_2 = 266$. Based on this information, we could use a 266KOhm resistor as R₁ and a 1 KOhm resistor as R₂.

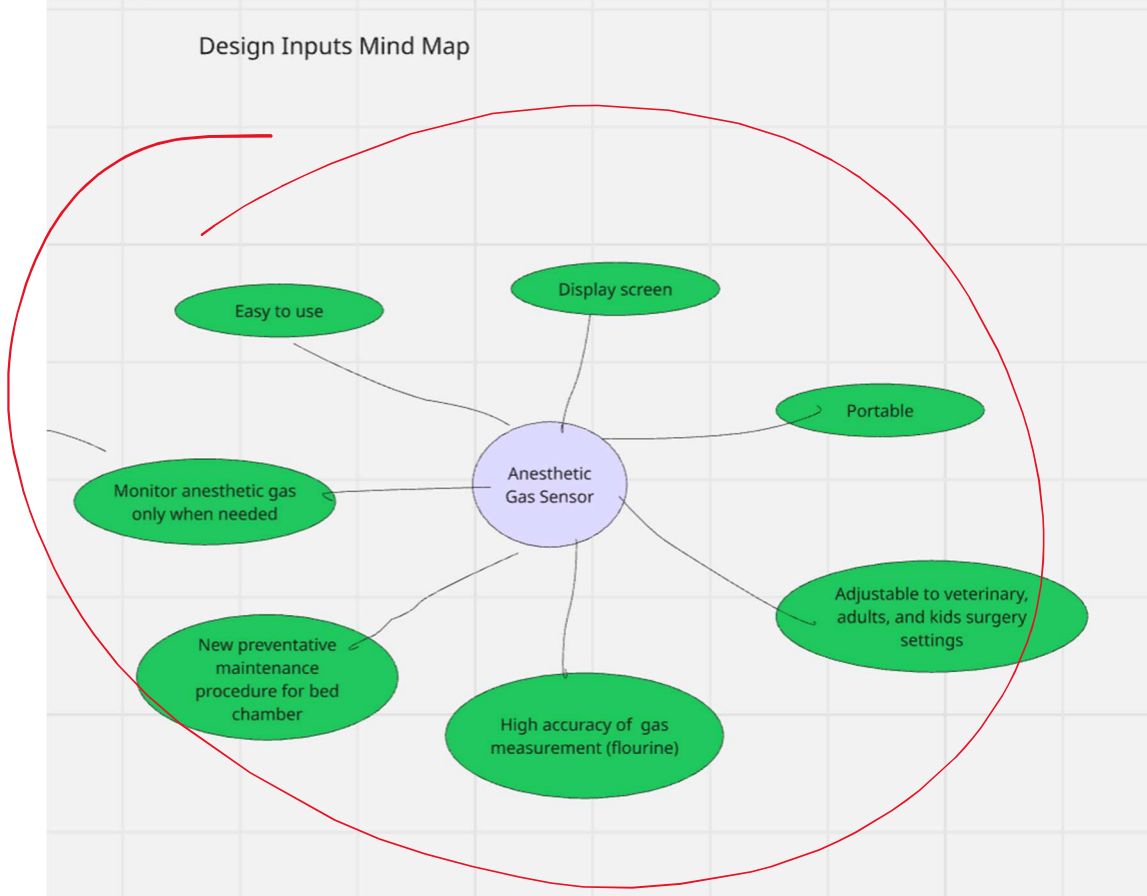
5. Bonus (2 pts): Solve the equations using the method or program of your choice and provide an example quantitative result to see if your intuition was correct from the question above.

I plugged in the values below:

$$\begin{aligned} R_1 &= 266k\Omega \\ R_2 &= 1k\Omega \\ V_{in} &= 9.35mV \\ A &= 1 + \frac{R_1}{R_2} \\ &= 1 + \frac{266k\Omega}{1k\Omega} \\ &= 1 + 266 \\ &= 267 \text{ gain} \end{aligned} \quad \begin{aligned} V_{out} &= V_{in} A \\ &= 0.00935 (267) \\ &= 2.50V \end{aligned}$$

When talking about this homework with my group mates we were able to start an interesting conversation about whether we would want to use non-dispersive infrared (NDIR) versus interference filter infrared (IFR) sensors. The solution I found above in this homework was an anesthetic waste gas sensor that was NDIR, while Kaayla based her calculations on a NIR sensor she found online. After doing further research I found that IFR sensors are a subset of NDIR sensors. NDIR sensors are the standard approach to anesthetic waste gas monitoring where a broad-spectrum of infrared light passes through a gas chamber and optical filters sort through these wavelengths before reaching the photodetector [4]. There are also Fourier-Transform Infrared (FTIR) sensors that could be a better solution, and they can capture all different gases at the same time while taking multiple wavelength measurements [4]. The NDIR sensors require multiple measurements for each gas and only takes in separate wavelength bands. The FTIR sensor is also said to calculate faster measurements and is portable, which matches our user needs [4]. This information will be helpful when we are choosing a sensor to build or looking for a sensor to use for our solution. During our stakeholder meeting it was also brought up that Dr. Bob has a sensor in mind. We have a meeting on Monday morning, so I will bring this information to him and see what his thoughts are.

Design Inputs Map



I started a design inputs mind map on our group's Miro board, so we can start getting a better picture of the project. Everything in green is a user need, our next step would be to add design inputs attached to the need, so we can further our understanding of the project. Once all members add to the map and some ideas are cut down, it will serve to be useful when brainstorming solutions and coming up with questions for the KidsOR team about their current solution. It will also help the group recenter if we get lost along the way in the design process.

References

- [1] K. Samad, Muhammad Saad Yousuf, H. Ullah, Syed Shabbir Ahmed, Khalid Maudood Siddiqui, and A. Latif, "Anesthesia and Its Environmental Impact: Approaches to Minimize Exposure to Anesthetic Gases and Reduce Waste," *Medical Gas Research*, vol. 15, no. 1, Jun. 2024, doi: 10.4103/mgr.medgasres-d-23-00059.
- [2] J. A. Barwise, L. J. Lancaster, D. Michaels, J. E. Pope, and J. M. Berry, "An Initial Evaluation of a Novel Anesthetic Scavenging Interface," *Anesthesia & Analgesia*, vol. 113, no. 5, pp. 1064–1067, Nov. 2011, doi: 10.1213/ane.0b013e31822c9a2c.

- [3] H.-B. Deng, F.-X. Li, Y.-H. Cai, and S.-Y. Xu, "Waste Anesthetic Gas Exposure and Strategies for Solution," *Journal of Anesthesia*, vol. 32, no. 2, pp. 269–282, Feb. 2018, doi: 10.1007/s00540-018-2448-1.
- [4] "Anesthetic Waste Gas Monitoring," Gasmet.com. Accessed: Sep. 11, 2026. [Online]. Available: <https://www.gasmet.com/applications/safety/anesthetic-waste-gas-monitoring/>